

4_HandlingNa's

February 11, 2025

```
[1]: import numpy as np
import pandas as pd
```

```
[2]: df = pd.read_csv('weather_data.csv')
df
```

```
[2]:
```

	day	temperature	windspeed	event
0	1/1/2017	32.0	6.0	Rain
1	1/4/2017	NaN	9.0	Sunny
2	1/5/2017	28.0	NaN	Snow
3	1/6/2017	NaN	7.0	NaN
4	1/7/2017	32.0	NaN	Rain
5	1/8/2017	NaN	NaN	Sunny
6	1/9/2017	NaN	NaN	NaN
7	1/10/2017	34.1	8.1	Cloudy
8	1/11/2017	40.0	12.0	Sunny

```
[3]: type(df.day[0])
```

```
[3]: str
```

0.1 changing data type of day column into timestamp

```
[4]: df = pd.read_csv('weather_data.csv', parse_dates = ['day'])
type(df.day[0])
```

```
[4]: pandas._libs.tslibs.timestamps.Timestamp
```

0.1.1 setting day column as index

```
[5]: df.set_index('day', inplace = True)
df
```

```
[5]:
```

	temperature	windspeed	event
day			
2017-01-01	32.0	6.0	Rain
2017-01-04	NaN	9.0	Sunny
2017-01-05	28.0	NaN	Snow

2017-01-06	NaN	7.0	NaN
2017-01-07	32.0	NaN	Rain
2017-01-08	NaN	NaN	Sunny
2017-01-09	NaN	NaN	NaN
2017-01-10	34.1	8.1	Cloudy
2017-01-11	40.0	12.0	Sunny

0.2 fillna()

```
[6]: df['temperature'].isna().value_counts()
```

```
[6]: temperature
False    5
True     4
Name: count, dtype: int64
```

```
[7]: #filling with 0's
```

```
[8]: df.fillna(0, inplace = True)
```

```
[9]: df
#all the values are filled with 0 which can skew the mean or median value
```

```
[9]:
```

	temperature	windspeed	event
day			
2017-01-01	32.0	6.0	Rain
2017-01-04	0.0	9.0	Sunny
2017-01-05	28.0	0.0	Snow
2017-01-06	0.0	7.0	0
2017-01-07	32.0	0.0	Rain
2017-01-08	0.0	0.0	Sunny
2017-01-09	0.0	0.0	0
2017-01-10	34.1	8.1	Cloudy
2017-01-11	40.0	12.0	Sunny

0.3 Filling Na Values with custom data

```
[10]: df = pd.read_csv('weather_data.csv', parse_dates = ['day'])
df
```

```
[10]:
```

	day	temperature	windspeed	event
0	2017-01-01	32.0	6.0	Rain
1	2017-01-04	NaN	9.0	Sunny
2	2017-01-05	28.0	NaN	Snow
3	2017-01-06	NaN	7.0	NaN
4	2017-01-07	32.0	NaN	Rain
5	2017-01-08	NaN	NaN	Sunny

6	2017-01-09	NaN	NaN	NaN
7	2017-01-10	34.1	8.1	Cloudy
8	2017-01-11	40.0	12.0	Sunny

```
[11]: df.fillna({
        'temperature' : df.temperature.mean(),
        'windspeed' : df.windspeed.mean(),
        'event' : 'No Event'
    })
```

```
[11]:
```

	day	temperature	windspeed	event
0	2017-01-01	32.00	6.00	Rain
1	2017-01-04	33.22	9.00	Sunny
2	2017-01-05	28.00	8.42	Snow
3	2017-01-06	33.22	7.00	No Event
4	2017-01-07	32.00	8.42	Rain
5	2017-01-08	33.22	8.42	Sunny
6	2017-01-09	33.22	8.42	No Event
7	2017-01-10	34.10	8.10	Cloudy
8	2017-01-11	40.00	12.00	Sunny

0.4 Forward Fill or Backward Fill

```
[12]: df.fillna(method = 'ffill') #limit = 1 will limit the number of fillings to one
```

```
/var/folders/hb/vhzb1bdj2dl394bb64f0y1nc0000gn/T/ipykernel_1195/1921065525.py:1:
FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a
future version. Use obj.ffill() or obj.bfill() instead.
```

```
df.fillna(method = 'ffill') #limit = 1 will limit the number of fillings to
one
```

```
[12]:
```

	day	temperature	windspeed	event
0	2017-01-01	32.0	6.0	Rain
1	2017-01-04	32.0	9.0	Sunny
2	2017-01-05	28.0	9.0	Snow
3	2017-01-06	28.0	7.0	Snow
4	2017-01-07	32.0	7.0	Rain
5	2017-01-08	32.0	7.0	Sunny
6	2017-01-09	32.0	7.0	Sunny
7	2017-01-10	34.1	8.1	Cloudy
8	2017-01-11	40.0	12.0	Sunny

```
[13]: df.fillna(method = 'bfill', axis = 'columns')
```

```
/var/folders/hb/vhzb1bdj2dl394bb64f0y1nc0000gn/T/ipykernel_1195/3664017729.py:1:
FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a
future version. Use obj.ffill() or obj.bfill() instead.
```

```
df.fillna(method = 'bfill', axis = 'columns')
```

```

/var/folders/hb/vhzb1bdj2dl394bb64f0y1nc0000gn/T/ipykernel_1195/3664017729.py:1:
FutureWarning: Downcasting object dtype arrays on .fillna, .ffill, .bfill is
deprecated and will change in a future version. Call
result.infer_objects(copy=False) instead. To opt-in to the future behavior, set
`pd.set_option('future.no_silent_downcasting', True)`
  df.fillna(method = 'bfill', axis = 'columns')

```

```

[13]:
      day temperature windspeed  event
0  2017-01-01 00:00:00      32.0     6.0   Rain
1  2017-01-04 00:00:00     9.0     9.0  Sunny
2  2017-01-05 00:00:00    28.0    Snow   Snow
3  2017-01-06 00:00:00     7.0     7.0   NaN
4  2017-01-07 00:00:00    32.0    Rain   Rain
5  2017-01-08 00:00:00  Sunny  Sunny  Sunny
6  2017-01-09 00:00:00   NaT   NaT   NaT
7  2017-01-10 00:00:00    34.1     8.1 Cloudy
8  2017-01-11 00:00:00    40.0    12.0  Sunny

```

```

[14]: df.interpolate()
      #fills na with mean value of the two values in between

```

```

/var/folders/hb/vhzb1bdj2dl394bb64f0y1nc0000gn/T/ipykernel_1195/3424417509.py:1:
FutureWarning: DataFrame.interpolate with object dtype is deprecated and will
raise in a future version. Call obj.infer_objects(copy=False) before
interpolating instead.
  df.interpolate()

```

```

[14]:
      day  temperature  windspeed  event
0  2017-01-01      32.0     6.000   Rain
1  2017-01-04      30.0     9.000  Sunny
2  2017-01-05      28.0     8.000   Snow
3  2017-01-06      30.0     7.000   NaN
4  2017-01-07      32.0     7.275   Rain
5  2017-01-08      32.7     7.550  Sunny
6  2017-01-09      33.4     7.825   NaN
7  2017-01-10      34.1     8.100 Cloudy
8  2017-01-11      40.0    12.000  Sunny

```

```

[15]: df

```

```

[15]:
      day  temperature  windspeed  event
0  2017-01-01      32.0     6.0   Rain
1  2017-01-04      NaN     9.0  Sunny
2  2017-01-05      28.0    NaN   Snow
3  2017-01-06      NaN     7.0   NaN
4  2017-01-07      32.0    NaN   Rain
5  2017-01-08      NaN    NaN  Sunny
6  2017-01-09      NaN    NaN   NaN

```

7	2017-01-10	34.1	8.1	Cloudy
8	2017-01-11	40.0	12.0	Sunny

```
[16]: df.dropna()
```

```
[16]:
```

	day	temperature	windspeed	event
0	2017-01-01	32.0	6.0	Rain
7	2017-01-10	34.1	8.1	Cloudy
8	2017-01-11	40.0	12.0	Sunny

```
[17]: df.dropna(how = 'all')
```

```
[17]:
```

	day	temperature	windspeed	event
0	2017-01-01	32.0	6.0	Rain
1	2017-01-04	NaN	9.0	Sunny
2	2017-01-05	28.0	NaN	Snow
3	2017-01-06	NaN	7.0	NaN
4	2017-01-07	32.0	NaN	Rain
5	2017-01-08	NaN	NaN	Sunny
6	2017-01-09	NaN	NaN	NaN
7	2017-01-10	34.1	8.1	Cloudy
8	2017-01-11	40.0	12.0	Sunny

```
[18]: df.dropna(thresh = 2)
```

```
[18]:
```

	day	temperature	windspeed	event
0	2017-01-01	32.0	6.0	Rain
1	2017-01-04	NaN	9.0	Sunny
2	2017-01-05	28.0	NaN	Snow
3	2017-01-06	NaN	7.0	NaN
4	2017-01-07	32.0	NaN	Rain
5	2017-01-08	NaN	NaN	Sunny
7	2017-01-10	34.1	8.1	Cloudy
8	2017-01-11	40.0	12.0	Sunny